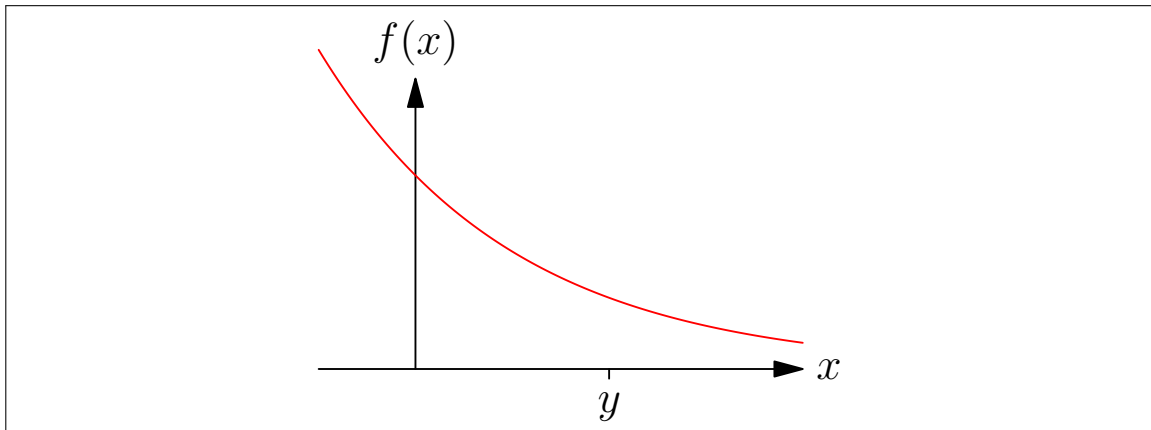


(b) Sketch the tangent line, $t(x)$, at the point y in the graph shown below. [1 mark]



1

(c) Starting from the inequality for a convex function

$$f(x) \geq f(y) + (x - y)f'(y) \quad (2)$$

consider the case $y = x + \epsilon$, then by Taylor expanding $f(x + \epsilon)$ and $f'(x + \epsilon)$ around x and keeping all terms up to order ϵ^2 show that for a convex function $f''(x) \geq 0$. [4 marks]

4

(d) Prove that $-\log(x)$ is convex-up for $x > 0$. [1 mark]

1

End of question 1

(a) $\frac{1}{4}$ (b) $\frac{1}{1}$ (c) $\frac{1}{4}$ (d) $\frac{1}{1}$ Total $\frac{10}{10}$

2

- (a) If $\|x\|$ is a proper norm use the triangular inequality ($\|x + y\| \leq \|x\| + \|y\|$), linearity of a norm ($\|ax\| = a\|x\|$) and the definition of convexity, to show that the norm is convex. [5 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

5

- (b) Consider a classification problem where $\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})$ is the probability that a learning machine with parameters $\boldsymbol{\theta}$ predicts that input $\mathbf{x}$ belongs to class $c \in \mathcal{C}$. Assume the training is stochastic so the probability of obtaining parameters $\boldsymbol{\theta}$ is $\rho(\boldsymbol{\theta})$. Let $\hat{m}_c(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\theta}}[\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})]$ be the output of the mean machine for class c . Assuming that for a data point $(\mathbf{x}, y)$, where y is a class label, we use a cross entropy loss

$$L(\mathbf{x}, y, \boldsymbol{\theta}) = - \sum_{c \in \mathcal{C}} \mathbb{I}[y = c] \log(\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})),$$

show that the expected loss over inputs and parameters can be written as the expected loss of the mean machine plus a second loss. Use Jensen's inequality ($\mathbb{E}[\log(X)] \leq \log(\mathbb{E}[X])$) to show the second term is positive. [5 marks]

5

End of question 2

(a) $\frac{5}{5}$ (b) $\frac{5}{5}$ Total $\frac{10}{10}$